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ATTENTION AND HYPERACTIVITY

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A Meta-Analysis of the Prevalence of ADHD in Youth Across Racial and Ethnic Groups

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ABSTRACT

Objective: To compare the prevalence of attention-deficit/hyperactivity disorder (ADHD) among Asian, Black, Latino, and White children and adolescents across studies while considering method of diagnosis.

Method: Studies were included if they reported the prevalence of ADHD among at least two different racial/ethnic groups in an unselected population. Studies were grouped based on the method of diagnosis; studies involving juvenile offenders were analyzed separately. A total of 9,019 articles were retrieved and 371 articles were evaluated. After excluding studies with overlapping datasets, 28 studies were included in the population-based meta-analysis, and five studies were included in the meta-analysis involving juvenile offenders.

Results: For studies asking caregivers if the child had received a diagnosis of ADHD (k = 23), the pooled prevalence of ADHD was lower for Latino (6.1%) and Asian (2.6%) youth than White youth (10.5%); the prevalence in Black youth (9.4%) was not significantly different from White youth. Among juvenile offenders (k = 5), the prevalence of ADHD was significantly lower in Black (19.1%) and Latino (17.0%) youth than in White youth (31.9%).

Conclusions: The present study finds some racial/ethnic differences in prevalence rates of ADHD. The study demonstrates that the type of sample and method for assessing ADHD diagnosis are important to consider.

Introduction

ADHD is a neurodevelopmental disorder associated with a range of negative long-term outcomes, including problems with academic performance, strained relationships with caregivers, peers, and teachers, poor health outcomes, and increased risk for criminal justice involvement (Egbert et al., 2018; Lifford et al., 2008; Rogers et al., 2015; Scholtens et al., 2013; von Polier et al., 2012). Several forms of treatment have been found to be effective in reducing ADHD symptoms (Coghill et al., 2023), including medication, behavioral interventions, and educational services (Wolraich et al., 2019). However, receipt of these treatments requires that children are appropriately identified and diagnosed. Several studies have documented disparities in ADHD diagnosis and treatment across racial and ethnic groups with White children being more likely to receive a diagnosis (Davis et al., 2021; Hall et al., 2023; Morgan et al., 2013). Other studies, including a recent meta-analysis (Cénat et al., 2024) have found little evidence of racial and ethnic differences in ADHD prevalence in youth (Danielson et al., 2024).

Understanding whether there are racial and ethnic disparities in ADHD diagnosis rates is important for several reasons. Disparities in prevalence may indicate inequities in sequence of steps necessary to obtain a diagnosis. This includes identifying children in need of assessment, which often depends on teacher and/or parent knowledge, access to resources for obtaining a diagnosis, and the receipt of comprehensive, evidence-based, unbiased assessments of ADHD. Identifying potential racial differences in prevalence may also highlight areas where targeted interventions and policy changes may be needed. Awareness of prevalence differences may also spur the development of culturally sensitive approaches to improving ADHD awareness and knowledge in families, and in reducing the stigma associated with an ADHD diagnosis (Slobodin & Masalha, 2020). This may ultimately improve outcomes for underserved groups.

When evaluating whether racial or ethnic disparities in ADHD diagnosis rates exist, it is important to consider how different studies determine the ADHD diagnosis of the study participants. In many studies,

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caregivers are asked whether their child has ever received a diagnosis of ADHD from a physician or mental health professional. This provides information about which children are receiving diagnoses within the context of existing educational and medical systems. However, it is unclear from this method whether differences in diagnosis rates between demographic groups are because of true differences in symptom levels, or because of differences in the identification of ADHD symptoms by caregivers and teachers, and/or differences in access to mental health services needed to obtain an assessment and diagnosis. For example, studies have found that caregivers from minoritized groups have less familiarity with or knowledge about ADHD and related treatments (Bussing et al., 2012; Coker et al., 2009), which may make these caregivers less likely to seek a diagnosis for their child. Additional issues such as potential bias in teachers' perceptions of children's behavior, cultural differences in perceptions of behavior, mistrust of institutions, and financial considerations may also differentially affect diagnosis rates across different racial and ethnic groups (Moody, 2016; Nasol et al., 2019).

To determine whether there are actual differences in ADHD symptoms levels between racial and ethnic groups, some studies have assessed ADHD symptoms within the context of the study to determine diagnosis rates. By screening every child in a classroom, for example, this method eliminates reliance on proper identification and access to ADHD assessments and may be more likely to accurately assess whether there are true differences in diagnosis rates between demographic groups.

One study used both methods for determining ADHD prevalence. Coker et al. (2016) assessed ADHD prevalence in a sample of 10th graders in two ways. First, they asked parents to rate their child's symptoms of ADHD and used this data to determine whether the child would meet criteria for ADHD or not. Then, parents were asked if a physician or health professional had ever told them that their child had ADHD. They found that 9% of White children meet criteria for ADHD based on parent-reported symptoms, but 19% reported having received a diagnosis from a professional. The opposite pattern was true for Black children, in which 13% met criteria for ADHD based on parent-reported symptoms, but only 10% reported having received a diagnosis. These results suggest that there may be true racial differences in actual ADHD symptoms levels (9% in White children compared to 13% in Black children), but there also may be differences in the ability to obtain a diagnosis of ADHD (19% of White children had received a diagnosis versus 10% of Black children).

The present study is novel because it uses meta-analysis to test whether these findings by Coker et al. (2016) hold true across a wide range of studies. We conducted separate meta-analyses for studies that conducted a diagnostic assessment of ADHD as part of the study and studies that relied on receipt of a diagnosis from a physician or mental-health professional. Receiving a diagnosis would require that a child previously be identified and then tested, which would necessarily require resources from the school and/or caregiver. In contrast, screening all youth for ADHD symptoms may provide a more objective measure of the true prevalence of ADHD in different racial and ethnic groups. We hypothesized that potential racial disparities in ADHD diagnosis would be larger in studies using caregiver reports of receiving a diagnosis, which rely on appropriate identification and diagnosis, compared to studies that directly assess ADHD symptoms in youth.

The present study expands on a recently published meta-analysis by Cénat et al. (2024) that examined racial and ethnic differences in ADHD prevalence. Cénat et al. (2024) did not find differences in prevalence between ethnic groups, or between Black and White children. They did find that the prevalence rate for Asian children was marginally lower than the prevalence rates for Black and White children. Because this study combined studies that used different methods for determining ADHD diagnoses, it is unclear how these results may change when examining studies with different methodologies separately. In addition, the meta-analysis by Cénat et al. (2024) included several studies that involved special or clinical populations such as youth with autism spectrum disorder or youth in the juvenile justice system, which are likely not representative of youth generally (for additional detail, see Supplementary Table S1). The present study includes only studies with population-based samples, and separates studies based on method of ADHD diagnosis. A separate analysis is also conducted for studies of children in the juvenile justice system.

Methods

Identification and Selection of Studies

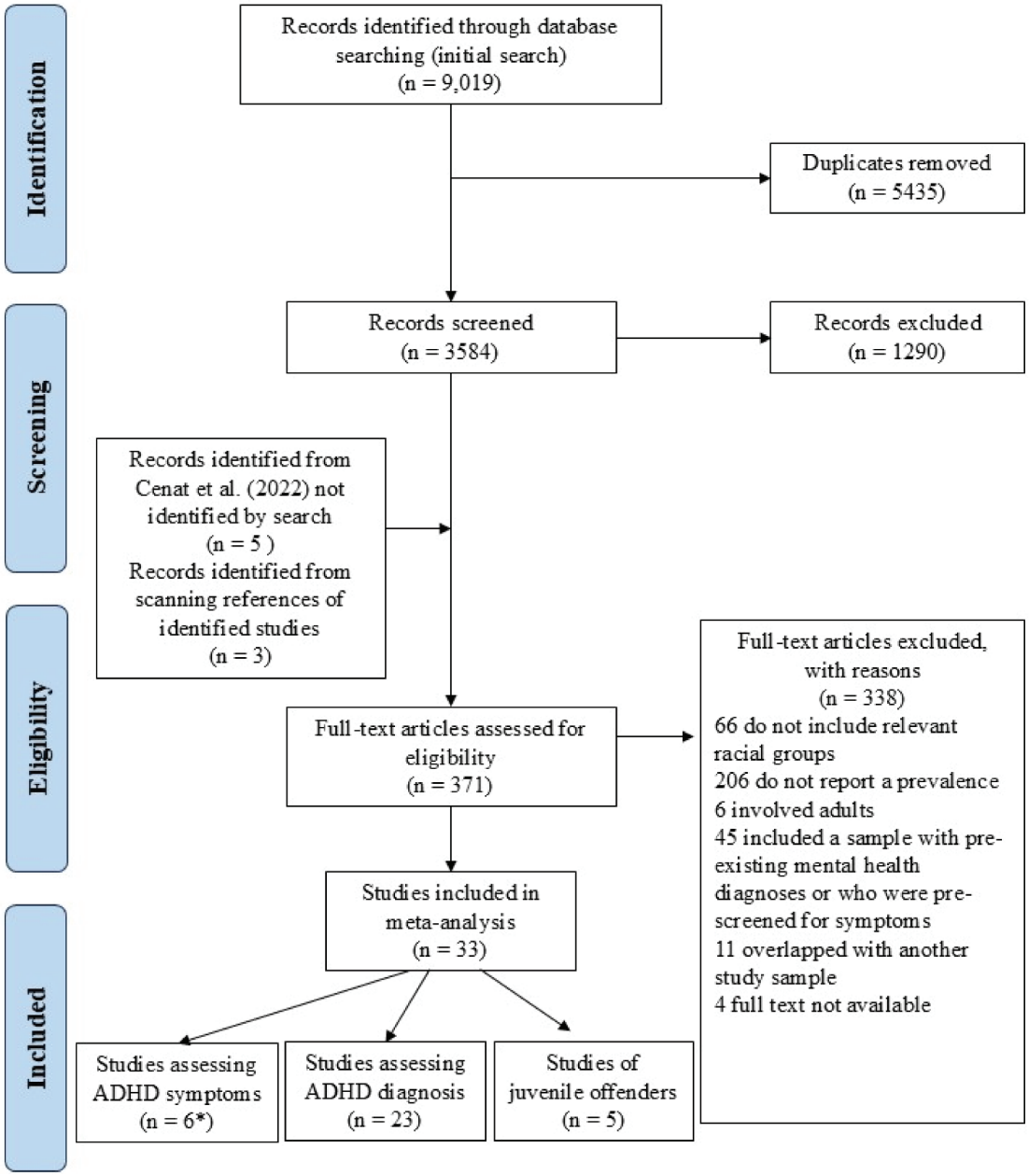
In the present study, we used the reporting guidelines from the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA; Moher et al., 2009). A literature search was conducted using the major search engines PsycINFO, Medline, PubMed, and Web of Science. The electronic search was

completed in May 2025. Search terms are available in the Supplementary Materials.

Following the search and retrieval process, identified articles were reviewed to determine whether they met the inclusion criteria described below (Figure 1). The initial search yielded a total of 9,019 articles. After removing duplicates, the first and third authors independently assessed the titles and abstracts of 3,584 articles. If one or both raters indicated that an article should be retained for full-text evaluation, the article was included in the next step. The 23 studies already identified in the meta-analysis by Cénat et al. (2024) were directly considered for inclusion.

Inclusions and Exclusion Criteria

Journal articles were included if they met the following criteria: (1) the article was peer reviewed; (2) the study reported empirical data on the prevalence of ADHD separately for at least two of the following racial/ethnic groups (Black, White, Latino, and Asian); (3) the study was conducted in countries where Black people live in a minority context and White people of European



*Coker et al. (2016) includes both symptom and diagnosis data and therefore is included in both

Figure 1. Study selection flow chart. *Coker et al. (2016) includes both symptom and diagnosis data and therefore is included in both.

ancestry live in a majority context (United States, Canada, European and Eastern European countries, Russia, Australia, New Zealand); (4) participants in the study were 0–18 years of age, (5) the study was published in English. We excluded studies that examined special populations such as youth with autism or other significant psychological disorders. We also excluded studies in which youth were pre-screened and selected for higher levels of ADHD symptoms, which would affect prevalence rates. We included studies involving juvenile offenders, but these studies were analyzed in a separate analysis. We limited studies to those published after 1994 aligning with the publication of DSM-IV.

Regarding ADHD diagnosis, we included studies that reported an ADHD diagnosis as a category (yes/no) and excluded studies that only reported mean levels of ADHD symptoms, as prevalence rates could not be determined from these studies. Within the studies that utilized a categorical ADHD diagnosis, we divided them into two groups. The first group consisted of studies that conducted a diagnostic assessment or in which researchers evaluated symptom levels to approximate diagnoses (e.g., parents reported on the child's ADHD symptoms); the second group consisted of studies that relied on parent- or self-reporting that the child had received a diagnosis from a physician or mental-health professional.

Coding

The two raters (first and third authors) identified 371 articles to be retained for full-text evaluation. Both raters independently reviewed the full text of these articles and coded them based on the inclusion and exclusion criteria. Each article that was excluded was coded based on the reason for exclusion (Figure 1). Each included article was categorized into one of four groups: (1) studies assessing ADHD symptoms to determine a diagnosis, (2) studies assessing receipt of diagnosis from a physician or mental health professional at any point in the child's lifetime, (3) studies assessing a current diagnosis from a physician or mental health professional, and (4) studies involving juvenile offenders. The first and third authors met to discuss discrepancies in the coding for inclusion/exclusion criteria, re-reviewed the articles, and came to a consensus about each article. From the 371 full-text articles, 39 met the inclusion criteria for the main meta-analysis, and five met criteria for the meta-analysis involving juvenile offenders. One of the five studies involving juvenile offenders included an age range from 10 to 21 (Harzke et al., 2012); due to the small number of studies involving juvenile offenders, we decided to include this study. Data in the Harzke et al. (2012) study were from youth committed to juvenile facilities, and 71.3% were 16 or younger at the time of commitment. Of the 39 studies meeting inclusion criteria for the main meta-analysis, many included data from the same overarching study or dataset (Supplementary Table S2). When multiple studies were identified from the same overarching study, studies were selected to maximize the number of participants included and to avoid overlapping data. Eleven studies were excluded due to overlapping data, resulting in a total of 28 studies included in the meta-analysis reflecting the general population of youth (Supplementary Table S2). Of these, six studies assessed ADHD symptoms in youth, and were therefore analyzed separately. The remaining 23 assessed lifetime or current ADHD diagnoses. No studies involving juvenile offenders utilized overlapping data. All of the included studies were conducted in the United States.

Table 1 depicts the characteristics of the included studies in the population-based meta-analysis. The table is organized based on whether the ADHD diagnosis was based on report of a diagnosis received or based on a threshold of reported symptoms. Table 1 also lists studies included in the meta-analysis of studies involving juvenile offenders.

Meta-Analysis

Analyses were performed on Comprehensive Meta Analysis Version 4. A random effects model was utilized to estimate overall pooled prevalence to account for expected variance across samples. For subgroup analyses, a mixed effects analysis was utilized. Heterogeneity was assessed in each meta-analysis using the I^2 statistic, Q, and prediction intervals (PIs). The I^2 statistic reflects the proportion of total variation in summary estimates across studies that is due to true heterogeneity rather than sampling error. PIs provide an estimate of the range within the true prevalence is expected to fall in comparable populations, reflecting the dispersion of estimates across studies. Sensitivity analyses were conducted in each meta-analysis to assess the potential for publication bias and robustness of summary estimates. These included the calculation of Egger's regression intercept, visual inspection of funnel plots, and a leave-one-out (one-study-removed) analysis to determine whether any one study had a disproportionate influence on the summary estimate. Given the use of a random-effects model, which tends to give relatively more weight to smaller studies (Borenstein et al., 2021), it was anticipated that smaller studies might disproportionately impact the summary estimate. Therefore, decisions about whether to

Table 1. Characteristics of the included studies for the meta-analysis.

Study	Age range/ mean	Total # of Black participants	# of Black participants with ADHD	Total # of White participants	# of White participants with ADHD	Total # of Latinx participants	# of Latinx participants with ADHD	Total # of Asian participants	# of Asian participants with ADHD
Studies Using Symptom Ratings									
Alegría et al. (2012)	13–17	1097	93	4257	344	758	67	n/a	n/a
Coker et al. (2016)	10th grade	1497	195	1039	87	1512	95	n/a	n/a
McConaughy et al. (2010)	6–12 (*M* = 8.2)	91	36	185	75	17	6	n/a	n/a
Inoue et al. (2019)	5–12	3124	184	7848	730	2360	167	n/a	n/a
Lavigne et al. (2009)	4 (*M* = 4.4)	133	28	433	39	162	27	n/a	n/a
Nolan et al. (2001)	3–18 (*M* = 9.8)	571	226	2194	312	n/a	n/a	n/a	n/a
Studies Using Reported Lifetime Diagnosis									
Bazargan et al. (2005)	0–18	84	16	n/a	n/a	100	15	n/a	n/a
Bozinovic et al. (2021)	3–17	3373	378	36,010	3926	6143	448	2522	73
Cervin (2023)	15	1594	245	588	123	809	100	n/a	n/a
Coker et al. (2016)	10th grade	1497	139	1039	191	1512	69	n/a	n/a
Collins and Cleary (2016) (2003 sample)	5–17	9001	729	41,680	4001	10,045	421	n/a	n/a
Collins and Cleary (2016) (2007 sample)	5–17	9197	1039	35,792	4116	11,931	764	n/a	n/a
Collins and Cleary (2016) (2011 sample)	5–17	8889	1138	34,043	4766	14,058	1082	n/a	n/a
Danielson et al. (2018)	2–17	2827	339	35,470	3334	4990	334	n/a	n/a
Garrison-Desany et al. (2022)	9–14 (median = 12)	1838	297	227	33	701	112	46	2
Hall et al. (2023)	2–17	367,733	216,772	394,393	32,630	1,568,484	88,125	47,149	696
Max et al. (2013)	4–15	2083	172	1622	148	2426	89		
Mennies et al. (2021)	9–10 (*M* = 9.9)	2481	638	8681	1891			814	140
Morgan et al. (2013)	K-8th grade	2226	67	9004	630	2818	124	n/a	n/a
Okano et al. (2019)	3–16 (*M* = 9.1)	1354	233	175	26	462	79	n/a	n/a
Pastor and Reuben (2005)	6–11	3562	221	11,287	858	5552	200	n/a	n/a
Shi et al. (2021)	4–18	14,792	682	173,082	8980	23,358	1034	15,183	365
Stevens et al. (2005)	3–18	4981	136	13,727	748	9094	177	n/a	n/a
Ullah and Kaelber (2021)	0–17	352,280	16,610	1,762,570	80,120	325,460	10,340	59,330	750
Wong and Landes (2022)	5–17	17,899	1917	68,112	7710	26,548	1598	5526	122
Studies Using Current Diagnosis									
Baumgardner et al. (2010)	5–17 (*M* = 12)	1704	180	15,384	2663	1180	111	272	10
Danielson et al. (2018)	2–17	2827	302	35,470	2979	4990	299	n/a	n/a
Davis et al. (2021)	6–17	43,407	5946	231,851	35,402	20,370	1238	3246	43
Getahun et al. (2013)	5–11	69,092	3998	160,510	13,446	340,347	12,466	46,022	1019
Okwori (2022)	3–17	1739	193	18,388	1981	3129	248		
Reyes et al. (2013)	5–17 (*M* = 11)	629	46	5984	247	420	7	303	4
Sikov et al. (2022)	6–11 (*M* = 8.6)	1141	155	111	20	353	47	n/a	n/a
Studies Involving Juvenile Offenders									
Baglivio et al. (2017)	0–18	5056	1638	2673	1147	1034	284	n/a	n/a
	0–18	723	213	522	144	102	27	n/a	n/a

(*Continued*)

Table 1. (Continued).

Study	Age range/ mean	Total # of Black participants	# of Black participants with ADHD	Total # of White participants	# of White participants with ADHD	Total # of Latinx participants	# of Latinx participants with ADHD	Total # of Asian participants	# of Asian participants with ADHD
El Sayed et al. (2016)	14–17 (M = 16.2)	413	25	208	38	293	14	n/a	n/a
Harzke et al. (2012)	10–21*	3574	683	2203	696	4692	568	n/a	n/a
Teplin et al. (2002)	10–18 (M = 14.9)	574	98	207	43	386	53	n/a	n/a
	10–18 (M = 14.9)	430	86	89	20	136	40	n/a	n/a
Vaughn et al. (2008)	M = 15.5	238	42	400	189	n/a	n/a	n/a	n/a

*The study by Harzke was included despite having an age range of up to 21. Over 70% of participants in the study were committed at age 16 or younger, and all were committed to a juvenile facility.

exclude individual studies based on sensitivity analyses were informed not just by their statistical impact, but also by a review of their methodological quality and any potential bias that might affect the reliability of the summary estimate.

The study consisted of three separate meta-analyses. The first analysis included studies that used ADHD symptom ratings in which researchers evaluated symptom levels to approximate diagnoses (k = 6). The second analysis included studies that used lifetime or current diagnosis of ADHD in which the caregiver was asked whether the child had ever received or currently had a diagnosis of ADHD from a mental health professional (k = 25). Separate sub-analyses for lifetime (k = 19) and current diagnoses (k = 7) can be found in the Supplementary Materials. The third analysis included studies that included juvenile offenders (k = 5). Overall estimates of ADHD prevalence are provided for each group of studies, along with prevalence estimates for each race/ethnicity within that group. Subgroup analyses were conducted to test the effect of race/ethnicity on the overall prevalence estimate for each group of studies. For groups with an adequate number of studies, meta-regression analyses using Method of Moments (also known as the DerSimonian and Laird method; Borenstein et al., 2021) were conducted to assess the unique impact of each racial/ethnic group on ADHD prevalence. Meta-regression analyses were conducted when at least 10 studies were available for a given moderator, consistent with commonly cited guidance aimed at ensuring sufficient power and stable estimates, though it is acknowledged that no universally agreed-upon cutoff exists (Borenstein et al., 2021; Higgins et al., 2024). This threshold was established a priori. In one subgroup (juvenile offenders), fewer than 10 studies were available (k = 5); however, an exploratory meta-regression was conducted post hoc to examine potential sources of heterogeneity within this unique sample population.

Results

Racial/Ethnic Group Comparison in ADHD in Studies Using Symptom Ratings

We first examined the prevalence rates of ADHD in different racial/ethnic groups for the studies that assessed ADHD symptoms to determine ADHD diagnosis (k = 6). Across these studies, the overall mean prevalence of ADHD was 14.0% (95% CI [9.3, 20.4]). However, there was a wide dispersion of effects, suggesting the prevalence in any true population could range from about 3% to 47% (PI [2.9, 47.0]). Heterogeneity statistics are as follows: Q = 561.52, p < .001, I^2 = 99.11. These figures indicate that the true effect does vary across studies and that over 99% of the variance in the current sample reflects variance in the true effect as opposed to due to sampling error.

For this group of studies, Egger's regression intercept, an indicator of publication bias, was not statistically significant. Visual analysis of the funnel plot (Supplementary Figure S1) revealed that smaller studies within this group did tend to produce somewhat larger effect sizes. However, most studies clustered near the combined effect estimate. The one study removed procedure indicated that removing the McConaughy et al. study would result in a smaller overall prevalence estimate (11.0%) suggesting it may have a disproportionate impact on the overall estimate. No other single study disproportionately impacted the overall estimate.

Despite this influence, the McConaughy et al. study was retained in the meta-analysis for several reasons. First, under a random-effects model, smaller studies may have greater influence on the summary estimate due to the way study weights are calculated. McConaughy et al. had a relatively small sample size compared to other studies using symptom rating scales. However, there was no evidence that this study introduced systematic bias; its methodology appeared sound and offered a reasonably unbiased estimate of ADHD prevalence. We considered it important to retain all methodologically reliable studies to ensure the summary estimate reflects the full range of available data and remains generalizable. Nonetheless, readers are encouraged to interpret not only the summary estimate but also the prediction intervals, which reflect the expected range of ADHD prevalence in a true population and help contextualize variability across studies.

Figure 2 depicts forest plots of the pooled prevalence rates of ADHD in Black, White, and Latino/a populations. Asian populations were not represented in this group of studies. The prevalence rate of ADHD among Black participants was 17.5% (95% CI [10.1, 26.7] PI [4.3, 49.7]), whereas it was 12.6% among White participants (95% CI [7.7, 19. 8] PI [3.0, 40.1]), and 11.2% among Latino/as (95% CI [6.3, 18.9] PI [2.5, 37.7]). Race/ethnicity was not a statistically significant moderator for ADHD prevalence among studies who used a symptom rating approach for determining diagnosis ($Q = 1.74$ $df = 2$, $p = .42$). These findings should be interpreted cautiously, as the results of subgroup analyses are impacted by the small number of studies, in addition to study-specific factors such as sample size and homogeneity of samples (Borenstein et al., 2021). Meta-regression analyses were not conducted for this group of studies (Table 2).

Racial/Ethnic Group Comparison in ADHD in Studies Using Lifetime or Current ADHD Diagnosis

When examining studies that used a diagnosis (lifetime or current; $k = 25$), the overall prevalence of ADHD was 9.2% (95% CI [7.7, 11.1], PI [3.4, 22.5]). Heterogeneity statistics are as follows: Q = 67,626.79 $df = 24$ $p < .001$ I2 = 99.97. These figures indicate that the true effect does vary across studies and that over 99% of the variance in the current sample reflects variance in the true effect as opposed to due to sampling error. Sensitivity analyses were conducted for each group of studies individually (i.e., studies using lifetime and current diagnosis) and are discussed in the Supplemental Materials. Briefly, there was no significant evidence of publication bias within these groups of studies.

Figure 3 depicts forest plots of the pooled prevalence rates of ADHD in Black, White, Latino/a, and Asian populations. The pooled prevalence rate of ADHD

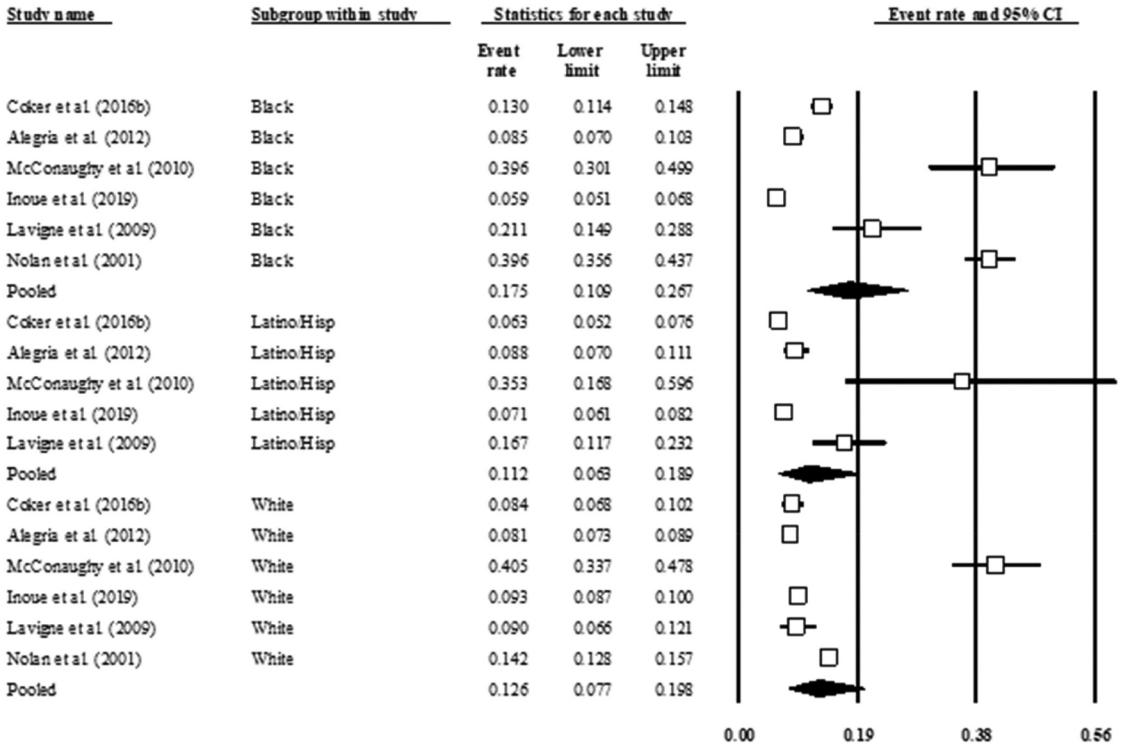


Figure 2. Forest plot for the pooled proportion of ADHD by racial group among studies using symptom ratings of ADHD.

Table 2. Results of meta-analysis, subgroup analyses, and meta-regression by diagnostic groups.

				Subgroup analyses			Meta-Regression	
Diagnostic Method (*k*)	Prevalence [95% CI]	Prediction Interval	I^2	Prevalence	Prediction Interval	*Q*	*Z*	R^2
Symptom Ratings (6)	14.0 [9.3, 20.4]	2.9–47.0	99.1	Black = 17.5 White = 12.6 Latino/a = 11.2	4.3–49.7 3.0–40.1 2.5–37.7	1.74	n/a	n/a
Lifetime Diagnosis (19)	9.2 [7.8, 10.8]	4.1–19.3	99.9	Asian = 2.9 Black = 9.2 White = 10.3 Latino/a = 6.1	1.0–7.9 3.6–21.6 4.1–23.8 2.3–14.9	43.17***	−5.93*** −0.77 reference group −3.54***	0.0
Current Diagnosis (7)	9.3 [5.7, 14.9]	1.4–42.3	99.9	Asian = 2.0 Black = 10.2 White = 10.7 Latino/a = 6.3	0.7–5.7 4.0–23.9 4.2–25.0 2.4–15.8	37.56***	−5.49*** −.022 reference group −2.24*	0.55
Juvenile Offenders (7)	25.5 [19.4, 32.6]	8.6–55.3	99.2	Black = 19.1 White = 31.9 Latino/a = 17.0	14.3–25.2 14.3–56.8 6.7–36.9	11.38**	−2.68** reference group −3.07**	0.35

* $p < .05$, ** $p < .01$, *** $p < .001$.

Note. Prevalence estimates, confidence intervals, and prediction intervals are reported in percentage units. Confidence intervals provide an indicator of the precision of the current estimate based on the samples included in the meta-analysis. Prediction intervals provide an indicator of the dispersion of the prevalence estimate in any true population. Asian participants were not represented in the group of studies who used symptom ratings or the juvenile offenders sample. White was used as the reference group when testing the unique effect of each race/ethnicity on ADHD prevalence in meta-regression analyses.

among Black participants was 9.4% (95% CI [7.7, 11.3] PI [3.4, 23.5]) and 10.5% among White participants (95% CI [8.6, 12.7] PI [3.8, 25.8]). Among Latino/a participants, the estimated prevalence was 6.1% (95% CI [4.9, 7.4] PI [2.1, 16.1]), and among Asian participants, 2.6% (95% CI [1.8, 3.6] PI [0.8, 7.5]). Subgroup analyses indicated there was a statistically significant effect of race/ethnicity among studies in this group ($Q = 61.39$ $df = 3$, $p < .001$). Meta-regression revealed a similar pattern that has been observed above: Asian ($Z = -7.21$, $p < .001$) and Hispanic/Latino/a ($Z = -3.76$, $p < .001$) groups were associated with statistically significantly lower ADHD prevalence rates (Table 2). The overall effect of race/ethnicity explained 0% of the variation in true effects ($R^2 = 0.0$).

Racial/Ethnic Group Comparison in ADHD in Juvenile Offenders

Finally, we examined the prevalence rates of ADHD in different racial/ethnic groups for the studies involving juvenile offenders ($k = 5$). Across the five studies, the overall prevalence of ADHD was 25.5% (95% CI [19.4, 32.6] PI [8.6, 55.3]). Heterogeneity statistics are as follows: $Q = 750.31$ $df = 6$, $p < .001$ $I2 = 99.2$. These figures indicate that the true effect does vary across studies and that over 99% of the variance in the current sample reflects variance in the true effect as opposed to due to sampling error. Egger's regression intercept was not statistically significant. Visual analysis of the funnel plot indicated symmetrical dispersion of smaller studies near the combined effect estimate (Supplementary Figure S11). The one-study-removed method indicated no one study disproportionally affected the overall prevalence estimate.

Figure 4 depicts forest plots of the pooled prevalence rates of ADHD in Black, White, Latino/a, and Asian populations. The pooled prevalence rate of ADHD among Black participants was 19.1% (95% CI [14.3, 25.2] PI [7.8, 40.0]), whereas it was 31.9% among White participants (95% CI [24.7, 39.9] PI [14.3, 56.8]), and 17.0% among Latino/as (95% CI [12.2, 23.3] PI [6.7, 36.9]). Asian populations were not represented in these studies. Subgroup analyses indicated there was a statistically significant effect of race/ethnicity among these studies ($Q = 11.38$ $df = 2$, $p < .01$). Meta-regression indicated Black and Hispanic/Latino/a groups were statistically significantly associated with lower estimates of ADHD prevalence ($Z = -2.68$, $p < .01$; $Z = -3.0$, $p < .01$, respectively; Table 2). The overall model testing the effect of race/ethnicity explained some but not all the variation in true effects ($Q = 1337.42$, $df = 17$, $p < .001$, $R2 = 0.35$). Of note, while the model accounted for a proportion of the between-study variance, the relatively small number of studies included (< 10) may limit the stability and generalizability of the findings. As such, results should be interpreted with caution and considered exploratory.

Discussion

The present meta-analysis evaluated differences in ADHD prevalence in racial and ethnic groups while

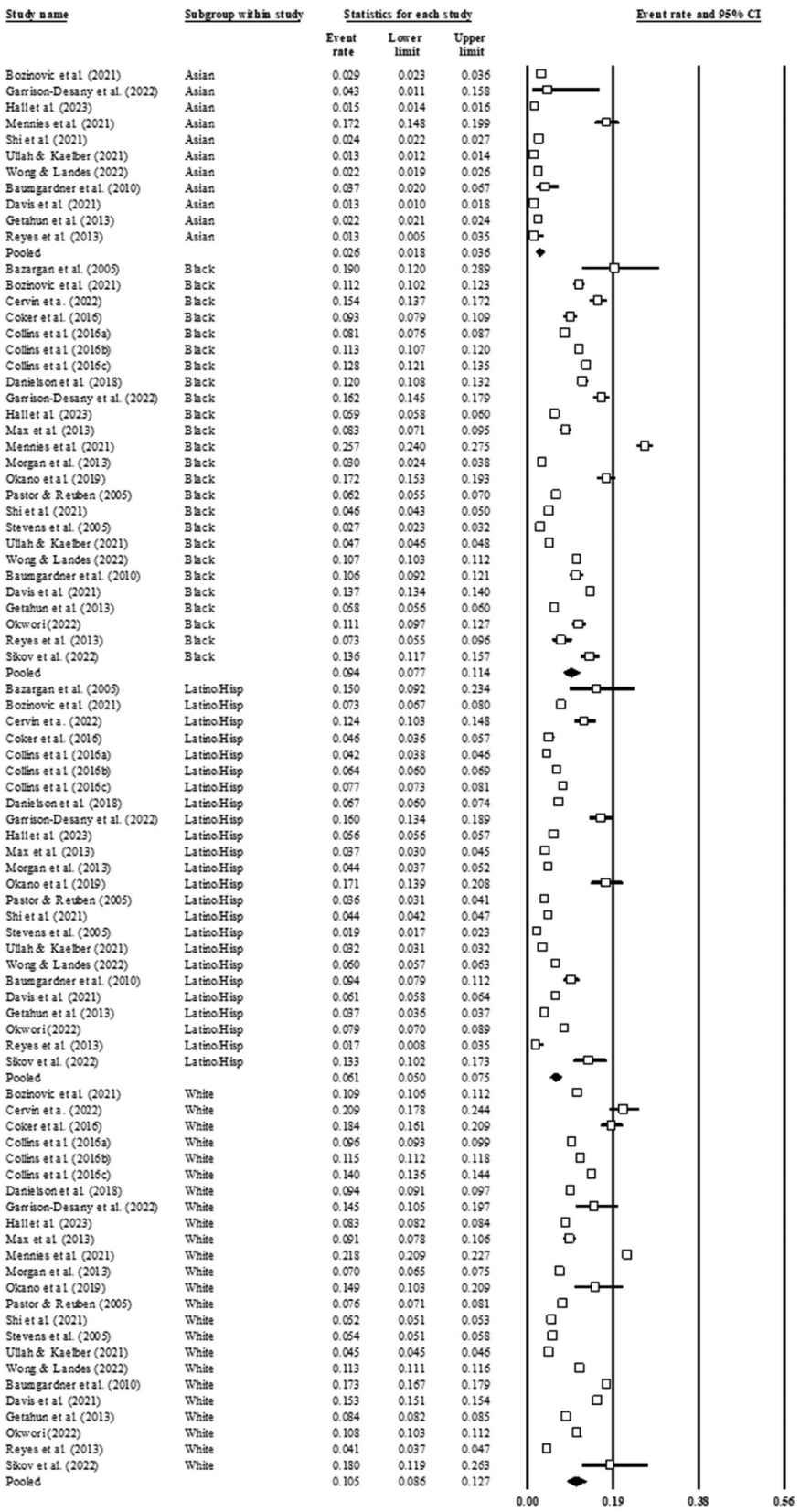


Figure 3. Forest plot for the pooled proportion of ADHD by racial group among studies using lifetime or current diagnosis of ADHD.

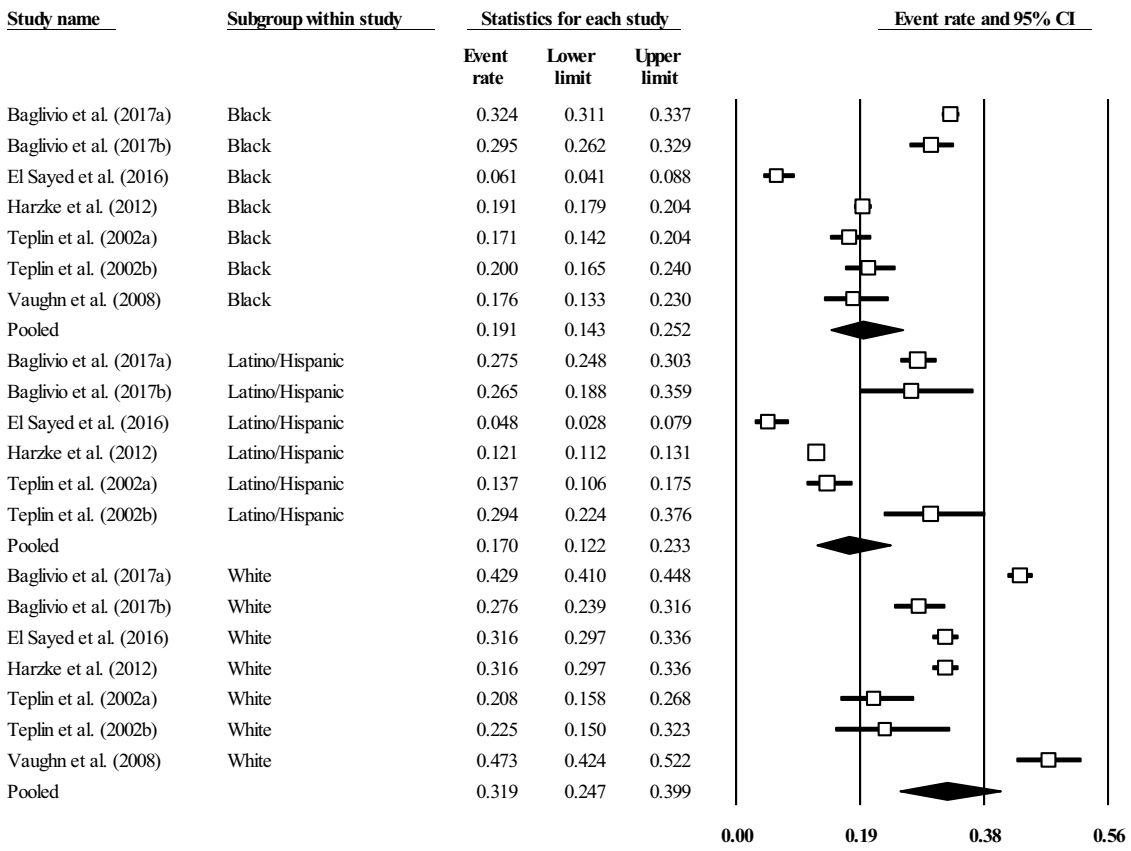


Figure 4. Forest plot for the pooled proportion of ADHD by racial group among studies of juvenile offenders.

considering the method of ADHD diagnosis. For the 25 studies in which caregivers were asked if their child had been diagnosed with ADHD by a physician or mental health professional (currently or ever), we found that Latino (6.1%) and Asian (2.6%) children had lower rates of diagnosis than White children (10.5%). These results contrast those of the recent meta-analysis by Cénat et al. (2024) who did not find lower prevalence of ADHD in Latino youth and found that the prevalence in Asian youth was only marginally lower. The differences in findings may be due to the types of populations included in the Cénat et al. (2024) meta-analysis (e.g., youth with autism, juvenile justice involved youth), or because of the inclusion of studies that involve different methods for determining ADHD diagnosis. The present study also included several additional studies that were not included in Cénat et al. (2024) such as the more recent large-scale study by Hall et al. (2023). The findings of the current study are consistent with recently published results from the 2022 National Survey of Children's Health (Danielson et al., 2024) which indicated that Hispanic/Latino children had a lower prevalence than non-Hispanic/Latino children, and that Asian children had a lower prevalence than White children.

These results suggest that it is important to investigate the reasons why rates of diagnosis may be lower in Latino and Asian communities. Several factors may contribute, including cultural differences in the perception of ADHD-like behaviors, language barriers, reduced access to mental health services generally, as well as a lack of access to and availability of culturally sensitive and linguistically appropriate mental health care services (Kapke et al., 2019; Feng et al., 2024). Haack et al. (2018) found that Latino families may be more likely to underestimate ADHD symptoms. Parental acculturation to the United States mainstream culture, stigmatized mental health attitudes, and beliefs that children's problems should be managed within the family unit also influence help-seeking in the Latino community (Kapke et al., 2019).

In Asian families, Feng et al. (2024) outline several factors that may contribute to lower rates of ADHD diagnosis. First, they note that the inattentive presentation of ADHD may be more common in Asian samples, which may be less likely to be identified as ADHD-related than the hyperactivity symptoms. They also suggest that the Model Minority Myth, which is the stereotype that all Asian Americans, through hard work and intelligence have succeeded in American society, may

influence the perceptions of teachers, school administrators, and healthcare providers, leading them to overlook ADHD symptoms. Additional factors include mental health stigma in Asian American communities; parent perception of ADHD as misbehavior rather than a neurodevelopmental disorder; and parent support for children's academic activities that may mask impairment. Culturally sensitive interventions with families and communities that focus on building relationships, address gaps in knowledge about ADHD symptoms, and providing information about opportunities for treatment may help to reduce the disparities in diagnosis rates in Latino and Asian communities.

Consistent with the findings of Cénat et al. (2024), for children with who had received a diagnosis of ADHD, we did not find differences in the prevalence rates for White (10.5%) and Black (9.4%) children. This is also consistent with the 2022 National Survey of Children's Health (Danielson et al., 2024), which did not show differences between prevalence rates in Black and White children. These findings suggest that previous concerns about the underdiagnosis of ADHD in Black children (Coker et al., 2016; Morgan et al., 2013) may no longer be present. However, our findings from studies using symptom ratings of ADHD suggest that underdiagnosis may still be an issue.

In the present study we distinguished between studies that asked caregivers about whether their child had received a diagnosis from studies that used symptom ratings of ADHD to assess prevalence. Whereas receiving a formal diagnosis requires that the child be identified as in need of assessment and that a caregiver is willing and able to seek the diagnosis, symptom ratings may be more objective indicators of prevalence. Only six studies used symptom ratings, so results should be interpreted with caution. However, we did find that the prevalence of ADHD was higher when using symptom ratings (14.0% when assessing symptoms versus 9.2% when asking about receipt of a diagnosis). This was true for each racial and ethnic group (17.5% compared to 9.4% in Black children, 12.6% compared to 10.5% in White children, 11.2% compared to 6.1% for Latino children (no studies reported symptom ratings in Asian youth)). These higher estimates when using symptom ratings may indicate that some children are being missed for diagnosis when universal screening of ADHD symptoms is not conducted. The differences appear to be particularly pronounced for Black and Latino children.

Of note, for studies using symptom ratings, there was a wide dispersion of effects, suggesting the true prevalence in any population could range from about 3% to 47% (PI [2.9, 47.0]). In contrast, the dispersion among studies using lifetime or current ADHD diagnosis was relatively smaller (PI [3.4, 22.5]). Greater dispersion may reflect methodological differences among studies using symptom ratings for ADHD identification (e.g., use of different rating scales), as well as variability in study-level characteristics such as sample size. Future research may assess how estimates of ADHD prevalence are influenced by these methodological factors. Nonetheless, the current results suggest the use of rating scales may improve ADHD detection. Underdiagnosis and thus a lack of treatment for ADHD may result in a wide range of negative outcomes, including poor academic performance, difficulties in social relationships, and increased likelihood of criminal justice involvement (Arnold et al., 2020; Retz et al., 2021). Various methods for improving the process of diagnosis and treatment have been proposed, including two-stage screening methods (Coghill et al., 2022), universally eliciting parental concerns about child behavior in primary care, and providing mental health education through community resources (Coker et al., 2016).

Only five studies involving juvenile offenders were identified, so results should be interpreted with caution. The rates of ADHD in juvenile offenders were much higher (25.5%) than in the general population of youth (9.2%), suggesting that these populations should be examined separately in meta-analyses. Further, across these five studies, we found that Black and Latino/Hispanic youth involved in the juvenile justice system had lower rates of ADHD than White youth (19.1% for Black youth, 17.0% for Latino youth, 31.9% for White youth; Asian samples were not included in the available studies). ADHD is associated with many factors that may increase risk for antisocial behavior, including deficits in executive functioning (e.g., impulsivity, poor self-control, altered reward processing, impaired emotion regulation), academic failure, and negative relationships with caregivers, teachers, and peers (Tamm et al., 2021). When undiagnosed and untreated, these factors are likely to increase risk for school suspensions and expulsions and involvement in the criminal justice system (Lettieri & Lewis, 2022). Underdiagnosis of ADHD in Black and Latino youth may contribute to the racial disparities present in the juvenile justice system.

In the study by Baglivio et al. (2017), Black and Latino juvenile offenders were diagnosed with ADHD less often but were *more likely* to be diagnosed with conduct disorder than White juvenile offenders, even when controlling for trauma, behavioral indicators, and criminal offending. One possibility is that among youth with more severe antisocial behavior, such as those in the juvenile justice system, clinicians may be more likely to overlook the ADHD diagnosis among Black and

Latino youth and focus on the conduct disorder diagnosis. Concerns have been raised that unconscious biases may affect the way clinicians judge and interpret behaviors based on race or ethnicity (Fadus et al., 2020). Addressing these potential biases is essential to ensure that vulnerable youth receive fair and objective evaluations and appropriate treatment.

The present study has several limitations. First, although race and ethnicity are separate constructs, based on the data reported in the studies, we were not able to include this distinction. Second, limited data was available for other racial and groups, including Asian, Middle Eastern, or Indigenous communities. Third, for the analyses involving lifetime and current diagnoses, the effect of race/ethnicity explained none of the variance in true effects. The reduced ability of the model's ability to account for a meaningful amount of variance in the overall effect may be related to a) lack of precision (i.e., high amount of variability) in the overall and between group estimates, b) relatively small number of studies in some subgroups, or some combination thereof. For these reasons, it is possible that the groups with smaller number of studies (e.g., juvenile offender groups) overestimate the true impact of race/ethnicity on ADHD prevalence in the available studies. Thus, results from these subgroups should be interpreted cautiously, and should be replicated when more studies are available to compare these groups with more precision and confidence.

An additional consideration when interpreting this study's findings is the significant degree of between study variability in ADHD prevalence, both in general and across racial/ethnic groups. The degree of variability can be assessed by focusing on the prediction intervals, which provides an indication of the extent of dispersion of the effect in any given population. For instance, when focusing on the group of studies using lifetime ADHD diagnosis, ADHD prevalence among White individuals could range from 4% to 24% in any given population. This suggests that the true rates of ADHD vary, and while the results of this study suggest that race/ethnicity may explain some of this variability, a significant amount of variability remains unexplained.

Even within the context of this study's limitations, the results provide evidence that race/ethnicity is statistically and clinically related to estimates of ADHD prevalence. In particular, Asian and Hispanic/Latino/a populations are consistently associated with lower rates of ADHD compared to White and Black populations. While Asian populations were not represented in the studies comprised of juvenile offenders, lower rates were observed among both Hispanic/Latino/a and Black youth compared to White youth. The present study improved on the previous meta-analysis of racial differences in the prevalence of ADHD by examining a more homogeneous population-based sample (e.g., excluding studies with comorbid autism spectrum disorder, and analyzing ADHD in juvenile offenders separately) as well as by considering the method of ADHD diagnosis assessment.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

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